

AYUSH RAJ

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EXPERIENCE

Software Developer Intern ↗

Horizon Mars Rover Team, CUSAT

📅 Nov 2023–Present 📍 Kochi, Kerala

- Developed and implemented a **URDF**-based robotic arm model with **TF** and **Gazebo** in ROS 2, including forward and inverse kinematics.
- Controlled a robotic arm using an Arduino Due (32-bit ARM) and servo motors in ROS 2, integrating real-time kinematics.
- Technologies used :** C++, ROS 2 (Humble), Linux, Arduino IDE

Embedded Intern ↗

Brainspired Labs Pvt. Ltd. 📍 OnSite, Kochi

📅 Jan 2025 – March 2025

- Integrated a dual 16-bit DAC with STM32H753ZI via **I2C**, analyzing output on a digital oscilloscope. Enabled **UART** communication in ESP32 with an Android app for frequency & voltage control.
- Integrated **CJMCU-1334** audio DAC to esp32-s3 using **I2S** protocol & gained hands-on experience with **AD5750** eval board.
- Technologies used :** STM32CUBE-IDE, Arduino IDE, Embedded C

PROJECTS

TurtleSim Project ↗



📅 Nov 2024

- Developed a Turtle Catcher game where turtles appear randomly, and the objective is to catch them all.
- Tech Stack:** Python, C++, ROS2 (Humble)

Mobile Robot Simulation ↗



📅 March 2025

- Developed an autonomous mobile robot with a robotic arm using URDF in ROS 2 Humble, integrating multiple TFs, links, and joints
- Implemented SLAM and Nav2 for autonomous navigation and successfully simulated the robot in Gazebo, demonstrating accurate kinematics and motion.
- Tech Stack:** ROS2 Humble, Rviz2, Xacro, URDF, Gazebo 11, SLAM, NAV2

Preventive AI ↗



📅 Sept 2024 - Oct 2024

- An AI platform combining **medical imaging**, **genomic data**, and **EHRs** for early cancer detection and personalized treatment insights.
- Tech Stack:** Python, Supervised ML, EJS, CSS, Node.js, Express.js, MongoDB Atlas, TensorFlow Serving

DIKSHA ↗



📅 Nov 2024

- Built a full stack application offering free educational tools, including **problem-solving resources**, an **NEP syllabus generator**, **AI chatbot**, **Doc QnA**, **Quiz**, and more.
- Tech Stack:** Bootstrap, Gemini API, Render, Node.js, Express.js, MongoDB Atlas, Vertex AI, LLMware

SOFT SKILLS

Leadership, Decision Making, Teamwork, Creativity

EDUCATION

Bachelor of Technology, Computer Science

Cochin University Of Science And Technology (CUSAT)

📍 Kochi, Kerala

📅 Oct 2022 – May 2026 (Expected)

Currently in **3rd Year (Sem-6)**- CGPA : 9.6

Academic Achievements

- Represented INDIA** at **European Rover Challenge, 2024, Poland** - Robotics Competition. Team name : Horizon (World Rank **18th**)
- Listed among **Top 10% Students** in the Batch.

POSITIONS OF RESPONSIBILITY

- Team Lead** @HORIZON- Mars Rover Team, CUSAT
- SIG AI Lead** @IEEE Execom'24- CS Branch Society,CUSAT

SKILLS

Languages

Java

Python

Javascript

C/C++

Embedded C

Frameworks and Libraries

React

OpenGL

ROS2

NodeJS

ExpressJS

RestAPI

AXIOS

Gemini API

Gazebo

SLAM

ROS2 NAV2

STM32CUBE - IDE

LLMware

Databases

MongoDB

SQL

Tools

Git/GitHub

Linux

Vertex AI

OOP

AI-ML

Proteus

ACHIEVEMENTS

- Won **5 Hackathons** (out of 10) conducted by **Google Development Group** ↗ (GDG) ,**Major League Hacking** ↗ & **Inovact** ↗.
- Solved **100+** programming problems across various platforms like **LeetCode**, **Hackerrank** etc.
- Jury Choice Award Winner** at AIC by **Accenture** & **Finalist** at **DevsHouse'24 Hackathon** by **VIT Chennai**

CERTIFICATIONS

ERC Space & Robotics Competition, AGH University Poland ↗

📅 Sep 2024

ROS2 Course Completion, Udemy ↗

📅 Nov 2024

ROS2 Level 2 Course Completion - Udemy ↗

📅 Feb 2025

Certificate of Appreciation (Tech. Team) - TEDxCUSAT ↗

📅 Sep 2024

Artificial Intelligence Fundamentals, IBM ↗

📅 June 2024